SMART INDIA HACKATHON 2024

SMART INDIA HACKATHON 2024

TITLE PAGE

- Problem Statement ID – **1660**
- Problem Statement- **Interactive gamified approach to ocean literacy**
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID- **3646**
- Team Name (Registered on portal)- **OKEANOS**

Okeanos

Interactive gamified approach to Ocean Literacy

SMART INDIA HACKATHON 2024

IDEA

- Decade of Ocean sciences initiative focuses on achieving the 17 Sustainable Development Goals (SDGs).
- A critical aspect of the agenda is transforming humanity's relationship with the oceans.
- INCOIS (Indian National Centre for Ocean Information Services) is designated as a UN Ocean Decade Collaborative Centre for the Indian Ocean Region.
- INCOIS is tasked with addressing the ten key challenges outlined by the UN Ocean Decade.

SOLUTION

IMMERSIVE OCEAN EXPLORATION GAME

Employs services of INCOIS and simulates sea conditions and expects the player to react accordingly.

Raises awareness to the common masses about the technology and services offered by INCOIS.

The solution simulates ocean challenges, rewarding correct decisions and penalizing mistakes, combining education with entertainment.

Okeanos

PROPOSED METHODOLOGY

SMART INDIA HACKATHON 2024

The main character in this game is a marine biologist who is on a mission to raise awareness about the ocean and help fix some of the modern day issues. These are the 4 islands we have created to encapsulate the challenges and services provided by the UN.

Dive into the fight against ocean plastic pollution and toxic algae. As a marine biologist, you'll clean up the waters and restore balance to the fragile ecosystem

INCOIS Services used:

- Water quality nowcast system
- ABIS

Face the challenge of **cleaning up hazardous sewage and dangerous oil spills threatening coastal waters and protect the ocean** and its inhabitants from contamination.

INCOIS Services used:

- Water quality nowcast system

As the marine biologist, you'll **navigate complex fishing zones, prevent overfishing**, and ensure the ocean remains a bountiful resource

INCOIS Services used:

- Potential Fishing Zones

As the marine biologist**, you'll plant new corals, heal damaged ecosystems, and bring life back to the underwater world through careful restoration efforts**.

INCOIS Services used:

- Coral Bleaching System

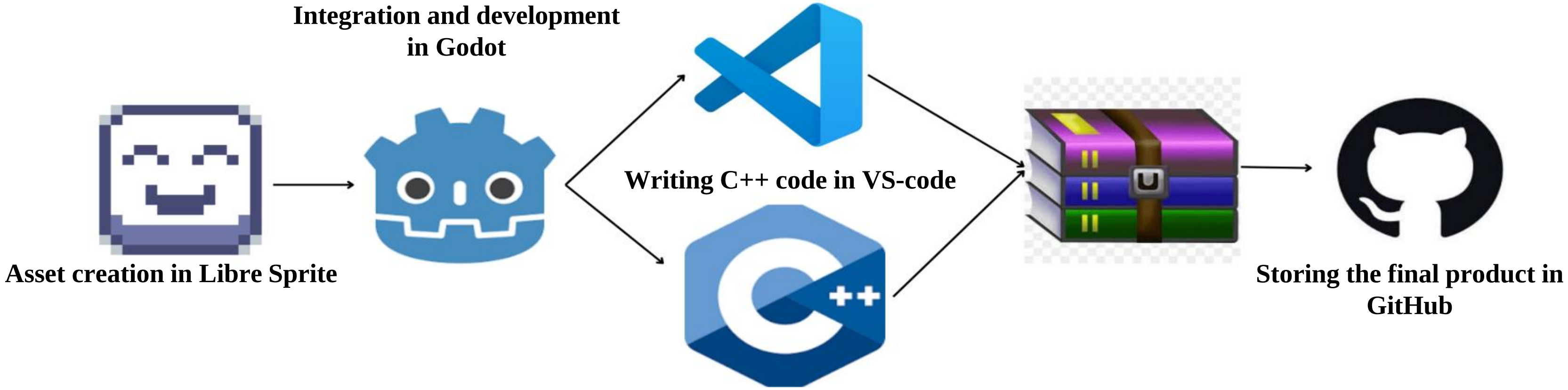
Storm surge warnings ,ocean state forecasts and early tsunami warning will be used when the player travels from island to island.

Okeanos

TECHNICAL APPROACH

SMART INDIA HACKATHON 2024

TECH STACK:



USER INTERFACE:

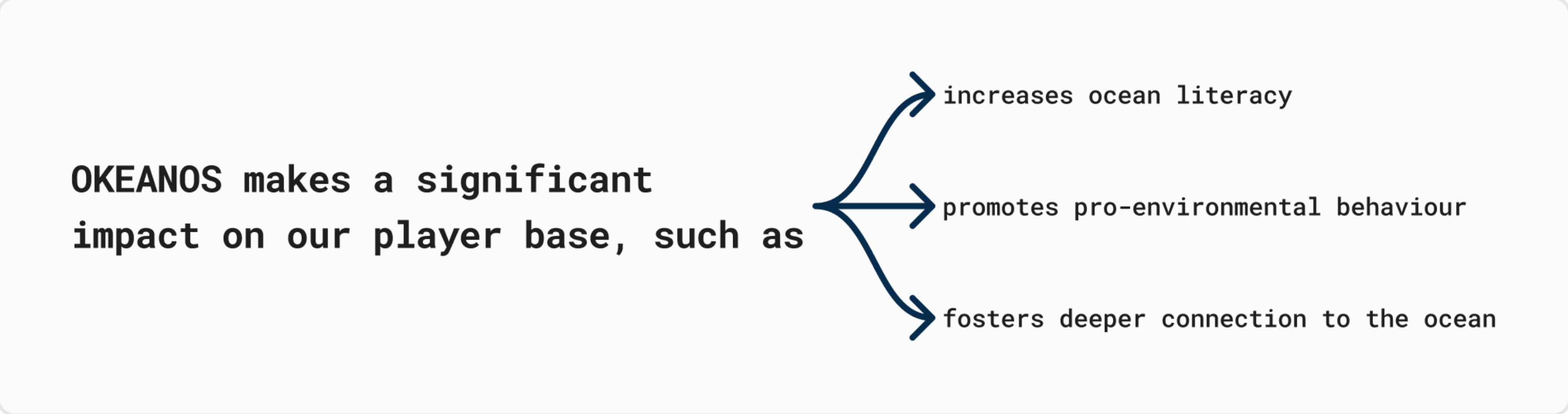
The user will be given a diving time 60-90 seconds depending on the level they're playing.

The following is the gaming interface which will be seen by the users.

Okeanos

FEASIBILITY AND IMPACTS

SMART INDIA HACKATHON 2024



1. **Feasibility** : The project is **technically and operationally feasible** with minimal financial barriers.
2. **Educational Impact**: It promises strong **educational benefits as well as societal benefits** and **aligns with the UN Decade of Ocean Science goals.**
3. **Challenges** : Issues include learning the Godot Engine, designing game mechanics, managing assets, balancing entertainment with education, and ensuring accessibility.
4. **Strategies** : Address challenges by learning Godot through tutorials, starting with simple mechanics, using iterative design, integrating educational content with rewards, and regular playtesting.
5. **Potential** : The game could be an **effective tool for raising ocean literacy awareness.**

Future Scope: Adding AR/VR elements making the end product more immersive and adding machine learning concepts that will add to the player experience.

Okeanos

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2024

- https://incois.gov.in/
- https://sih.gov.in/
- https://oceandecade.org/challenges/
- https://www.outreachgames.org/OceanProtector/main.html
- [1] M. Ariana, I. N. Suyasa, and D. Simbolon, "Remote sensing for assessing the potential anchovy fishing ground in the Pesisir Selatan Regency, West Sumatra, Indonesia," AACL Bioflux, vol. 13, no. 4, pp. 2273–2282, 2020. Available: http://www.bioflux.com.ro/aacl.
- [2] S. M. Chilaule, X. V. Macuacua, A. P. Mabica, N. A. Miranda, H. d. S. Pereira, E. S. Gudo, T. Marrufo, S. García-López, and M. Lopes, "Natural Disasters' Impact on Water Quality and Public Health: A Case Study of the Cyclonic Season (2019–2023)," *Pollutants*, vol. 4, pp. 212–230, 2024. Available: https://doi.org/10.3390/pollutants4020014.
- [3] N. Ya'acob, N. N. S. N. Dzulkefli, M. A. A. Aziz, A. L. Yusof, and R. Umar, "A review on features and methods of potential fishing zone," *International Journal of Electrical and Computer Engineering (IJECE)*, vol. 14, no. 3, pp. 2508-2521, June 2024, doi: 10.11591/ijece.v14i3.pp2508-2521.
- [4] E. McKinley, D. Burdon, and R. J. Shellock, "The evolution of ocean literacy: A new framework for the United Nations Ocean Decade and beyond," Marine Pollution Bulletin, vol. 186, 2023, Art. no. 114467. Available: https://doi.org/10.1016/j.marpolbul.2022.114467.
- [5] J. B. Wadler, J. E. Rudzin, B. Jaimes de la Cruz, J. Chen, M. Fischer, G. Chen, et al., "A review of recent research progress on the effect of external influences on tropical cyclone intensity change," Tropical Cyclone Research and Review, vol. 12, pp. 200-215, 2023, doi: 10.1016/j.tcrr.2023.09.001.